

Music Genre Classification Based on CNN

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Abstract—This paper implements an automatic music genre classification system based on Convolutional Neural Networks (CNN). The system extracts Log-Mel spectrograms as features from raw audio and constructs a deep model with multiple convolutional, pooling, batch normalization, and fully connected layers. By training and evaluating on the public GTZAN dataset, the model achieves 77% accuracy at the segment level and improves to 85% at the song level through a majority voting strategy. Experimental results show that the CNN model with Log-Mel features significantly outperforms classifiers based on traditional handcrafted features such as MFCC and achieves performance comparable to current state-of-the-art methods. This paper details the data preprocessing, feature extraction, model architecture, training strategies, and evaluation methods, and provides multi-dimensional visualization and analysis of the results.

Index Terms—Music Classification, Convolutional Neural Network, Log-Mel, Deep Learning, Audio Signal Processing

I. INTRODUCTION

Convolutional Neural Networks (CNN), as a powerful deep learning model, have been widely applied in computer vision, speech recognition, natural language processing, and other fields since their breakthrough in image recognition in 2012. Their core advantage lies in the ability to automatically learn hierarchical feature representations from raw data, thereby avoiding the cumbersome and domain-knowledge-dependent handcrafted feature engineering in traditional methods.

In audio signal processing, especially in Music Information Retrieval (MIR), traditional classification methods typically rely on handcrafted features such as Mel-Frequency Cepstral Coefficients (MFCC), spectral centroid, and zero-crossing rate, combined with classifiers like Gaussian Mixture Models (GMM) and Support Vector Machines (SVM). However, these methods have limitations in capturing complex, high-order nonlinear patterns and spatiotemporal structures in music audio.

In recent years, research applying CNN to audio classification has increased, achieving success in environmental sound classification, speech recognition, and other areas. These works typically convert audio signals into time-frequency representations (such as spectrograms), treating them as two-dimensional "images" to leverage CNN's powerful spatial feature extraction capability. The Log-Mel spectrogram, which aligns with human auditory characteristics and effectively represents the texture information of sound, has become the preferred input feature for audio CNN models. Although existing studies have validated the effectiveness of this paradigm,

how to design a clearly structured, parameter-efficient, and reproducible customized CNN model on standard datasets and conduct a complete analysis of its feature learning and decision-making process remains a practical topic worthy of further exploration.

This paper aims to implement music genre classification on the GTZAN dataset: to achieve effective automatic music genre classification using a deep model based on Log-Mel and CNN on a relatively limited public music dataset. To achieve this goal, we designed and implemented a deep CNN model comprising multiple convolutional blocks, global average pooling, and regularization components centered on the Log-Mel spectrogram feature. The core of this research is to verify the effectiveness of this customized model and systematically demonstrate the entire process from feature extraction to classification decision through a complete pipeline implementation and comprehensive experimental evaluation.

II. RELATED WORK

Music automatic classification has long relied on handcrafted features and classical machine learning methods. Early studies commonly extracted features such as Mel-Frequency Cepstral Coefficients and spectral centroid from audio and employed Gaussian Mixture Models or Support Vector Machines for classification. Such methods depend on the quality of feature engineering and have limitations in capturing complex nonlinear patterns in music.

In recent years, deep learning, especially Convolutional Neural Networks, has achieved remarkable success in the audio domain. Research confirms that converting audio into time-frequency representations (such as spectrograms) and treating them as images allows CNNs to automatically learn effective feature representations. Piczak et al. first applied CNN to environmental sound classification, demonstrating its potential for audio tasks. In the music domain, Dieleman et al. used CNN for end-to-end learning of music audio, achieving results comparable to traditional methods.

Notably, the Log-Mel spectrogram has become the mainstream input feature for audio CNN models due to its alignment with human auditory characteristics. Meanwhile, regularization techniques such as **Global Average Pooling** and **Dropout** are widely used to improve the generalization ability of CNNs in audio tasks by reducing parameters and preventing overfitting, thereby enhancing model robustness.

This work follows the above research tradition, with the core focus on **specifically implementing and analyzing a customized model based on this paradigm**. Compared to studies using large pre-trained networks or highly complex structures, this paper emphasizes providing an example with **clear module design, complete training pipeline, and comprehensive evaluation dimensions**. It details how a CNN with three convolutional blocks, global average pooling, and Dropout layers extracts features layer-by-layer from Log-Mel spectrograms and completes classification, aiming to provide a clear, reproducible reference for model design and analysis in this field.

III. SYSTEM ARCHITECTURE

A. Overall Architecture

The overall workflow of the system is shown in Fig. 1, mainly consisting of four stages: First, data preprocessing splits each 30-second song into multiple 5-second segments. Then, for each audio segment, its Log-Mel spectrogram is extracted, standardized, and padded/truncated to form a fixed-size two-dimensional matrix as the input feature. Finally, the processed spectrograms are used to train a customized CNN model. Model performance is evaluated on the test set at two levels: segment-level and song-level (based on majority voting).

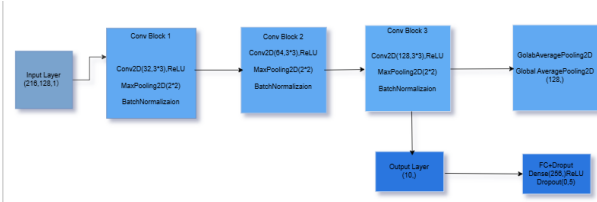


Fig. 1. Overall system workflow.

B. Data Preprocessing and Feature Extraction

The experiment uses the GTZAN dataset, which includes 1000 audio clips (30 seconds each) across 10 genres, with 100 clips per genre. To increase data volume and model robustness, each song is uniformly divided into 6 non-overlapping 5-second segments, expanding the dataset to 6000 samples.

For feature extraction, audio is loaded at a 22.05 kHz sampling rate, and its Log-Mel spectrogram is computed and converted to logarithmic scale (dB). Specific parameters are: FFT window length 2048, hop length 512, number of Mel filters 128. Finally, each 5-second segment is converted into a matrix of shape (time frames, 128). To ensure consistent input size, a maximum number of time frames (MAX_FRAMES=216) is set, with zero-padding for shorter segments and truncation for longer ones. This Log-Mel representation provides the distribution of sound energy along time and Mel frequency axes, making it an ideal input for CNN models.

C. Model Architecture

The CNN model is defined in model.py, built using the Sequential Keras API, and its architecture is shown in Fig. 1. The input layer receives single-channel Log-Mel spectrograms of shape (216, 128, 1). The feature extraction module consists of three sequentially stacked convolutional blocks to achieve feature abstraction from low-level to high-level:

- The first convolutional block uses 32 3x3 convolutional kernels for feature extraction, followed by ReLU activation, 2x2 max pooling downsampling, and batch normalization.
- The second convolutional block increases the number of filters to 64, also using 3x3 convolutional kernels, ReLU activation, 2x2 max pooling, and batch normalization to integrate more complex patterns.
- The third convolutional block further expands the filters to 128, performing the same sequence of convolution, activation, pooling, and batch normalization to output high-level semantic features.

This design systematically extracts high-level semantic features related to rhythm contours, harmonic structures, and even genre styles from low-level time-frequency patterns such as timbre and instrument onsets by gradually increasing the number of filters and reducing the spatial dimensions of feature maps.

The classification module first compresses the three-dimensional feature maps into a one-dimensional feature vector through a global average pooling layer. This design replaces the traditional flattening layer, significantly reducing the number of parameters while enhancing the model's robustness to minor variations in the time-frequency positions of features. Subsequently, a fully connected layer with 256 neurons and ReLU activation performs nonlinear mapping and combination of these features. To suppress overfitting, a Dropout layer with a rate of 0.5 is applied after this layer, forcing the network to learn more robust feature representations by randomly masking some neurons during training. Finally, the output layer consists of 10 neurons corresponding to the ten music genres, and the Softmax activation function outputs a normalized probability distribution to complete the classification decision.

D. Training and Data Augmentation

To enhance model generalization, SpecAugment data augmentation is applied to the input Log-Mel spectrograms during the training phase. As shown in Fig. 2, this technique randomly masks the spectrogram in the frequency domain (horizontal direction) and time domain (vertical direction), generating augmented samples with local information loss. This forces the convolutional neural network to learn more robust feature representations that do not rely on specific time-frequency regions, which is a key strategy to prevent model overfitting.

IV. EXPERIMENTS AND RESULTS

A. Dataset and Experimental Setup

Dataset: This study uses the public GTZAN music genre dataset. It includes 10 genres: Blues, Classical, Country, Disco, Hip-hop, Jazz, Metal, Pop, Reggae, and Rock. Fig. 3 shows the class distribution of the GTZAN dataset. It can be seen

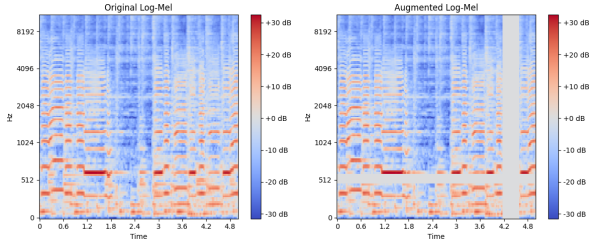


Fig. 2. SpecAugment data augmentation applied to Log-Mel spectrogram.

that the dataset includes 10 music genres, each containing 100 songs, and the class distribution is completely balanced. This balanced distribution helps avoid bias caused by class imbalance during model training, providing fair experimental conditions for subsequent model learning and performance evaluation.

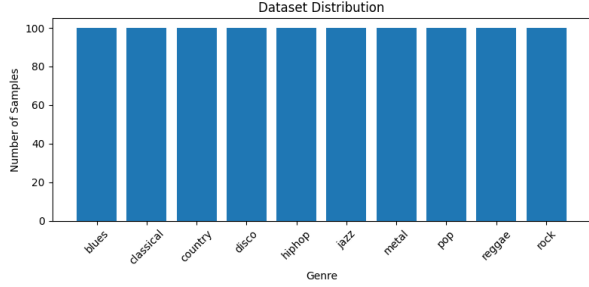


Fig. 3. Class distribution of the GTZAN dataset.

Experimental Setup: To increase data volume and model robustness, each 30-second audio clip is uniformly divided into 6 non-overlapping 5-second segments. Model training uses the Adam optimizer, loss function is categorical cross-entropy with label smoothing (coefficient 0.1), batch size 32, for a total of 40 epochs. Evaluation employs a dual standard: segment-level evaluation (evaluating 6000 independent segments) and song-level evaluation (for the 1000 original songs, using majority voting on predictions from 6 segments to derive the final label). SpecAugment data augmentation is applied during training, along with regularization techniques such as Dropout and batch normalization to prevent overfitting.

B. Model Configuration

The proposed CNN model is implemented using the Keras Sequential API. The model is trained for 40 epochs with a batch size of 32, using the Adam optimizer. To prevent overfitting and improve generalization, we adopt a combination strategy:

1. **Loss Function:** Categorical cross-entropy with label smoothing (smoothing factor=0.1).
2. **Regularization:** Dropout (dropout rate=0.5) applied after the fully connected layer.
3. **Internal Normalization:** Batch normalization after each convolutional block to stabilize and accelerate training.

C. Evaluation Strategy

This paper adopts a dual evaluation strategy to comprehensively assess model performance: segment-level evaluation and song-level evaluation. Segment-level evaluation assesses the prediction results of each independent 5-second audio segment, measuring the model’s ability to discriminate short-term music content. Song-level evaluation uses a majority voting strategy on the predictions of 6 segments belonging to the same original 30-second song to determine the final genre label for the entire song. This method better aligns with real-world application scenarios of classifying entire songs and can smooth local prediction errors at the segment level.

Fig. 4 shows the comparison of recognition accuracy between segment-level and song-level in the music genre classification task. It is clear from the figure that song-level accuracy is significantly higher than segment-level. This indicates that in song-level analysis, the model can utilize more complete contextual information, leading to more stable and accurate judgments. The relatively lower segment-level accuracy is due to local features of audio segments being insufficient to represent the overall category, or variability and noise within segments affecting model performance. This result aligns with our expectations and is consistent with common observations in the audio/music processing field: global information is crucial for high-level semantic tasks such as song classification. The experimental results provide a basis for subsequent model design, suggesting that when dealing with segment-level tasks, context modeling or attention mechanisms may need to be introduced to compensate for the lack of local information.

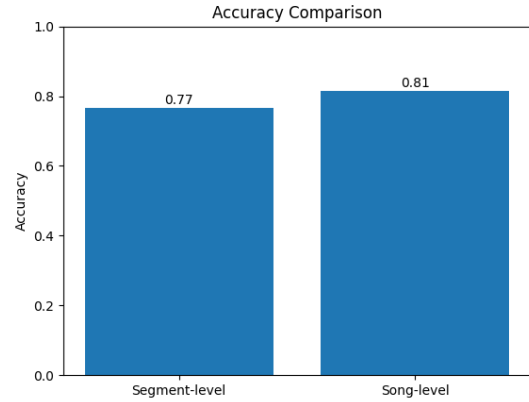


Fig. 4. Comparison of segment-level and song-level accuracy.

D. Results

To comprehensively evaluate the performance of the model in the music genre classification task, we analyze it from both segment-level and song-level granularities. We quantify model performance and explore its internal patterns through confusion matrices, F1-scores per category, and detailed classification reports.

Song-level classification takes the entire song as the input unit, enabling the utilization of complete temporal context information. Fig. 5 shows the confusion matrix of the model across 10 genres, intuitively presenting classification details. It can be seen that the model achieves very high recognition accuracy for genres with distinct styles such as Blues, Classical, and Metal (all $\geq 95\%$). The main confusion occurs between genres with similar styles: Blues and Rock misclassify each other 5 times each; nearly half of Country samples are misclassified as Blues (6 times) or Pop (3 times); Reggae also exhibits some confusion with Blues and Hip-hop. This indicates that the model can effectively capture global style features but still has room for improvement in distinguishing between genres that share similar instruments, harmonies, or rhythms.

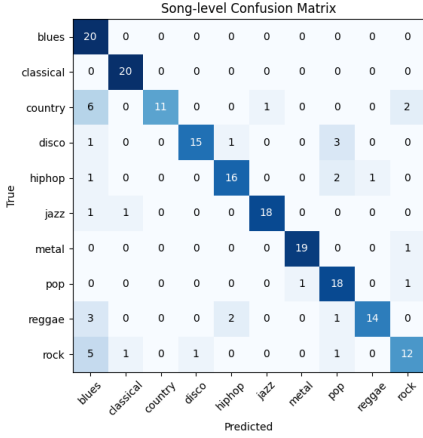


Fig. 5. Confusion matrix for song-level classification.

To further quantify performance, Fig. 6 shows the F1-score distribution across genres. The model performs excellently on most genres, with Blues, Classical, Jazz, Metal, and Pop all achieving F1-scores above 0.90. Country (0.71) and Rock (0.65) are the two relatively weaker categories, consistent with observations in the confusion matrix. The macro-average F1-score for the song-level task reaches 0.85, indicating the overall effectiveness of the model at this granularity.

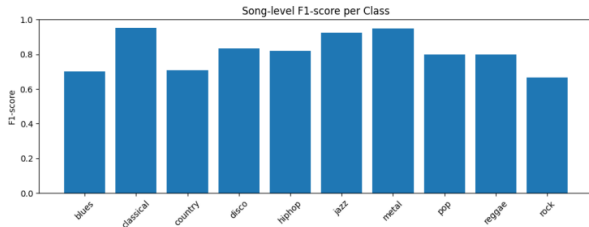


Fig. 6. F1-score distribution across genres for song-level classification.

As shown in Table I, from segment-level to song-level, the overall accuracy of the model improves from 0.77 to 0.85, and the macro-average F1 score similarly increases from 0.77 to 0.85. This clearly demonstrates that utilizing complete

contextual information at the song level can significantly improve the accuracy of music genre classification, validating the effectiveness of long-term feature modeling. Comparative analysis reveals differences in how genres respond to song context: Blues, Pop, and Reggae benefit most significantly, with their F1 scores markedly improving, and the precision-recall imbalance existing at the segment level is alleviated at the song level; most genres (Classical, Jazz, Metal, Hip-hop, Disco) maintain stable and excellent performance, indicating their acoustic features already have high discriminability at the segment level; while Rock shows improvement, it remains limited by confusion with Blues. The only exception is Country, whose song-level recall increases accompanied by a significant drop in precision, reflecting that the stylistic boundaries of this genre become blurred in long-term contexts, posing a new classification challenge.

TABLE I
PERFORMANCE COMPARISON BETWEEN SEGMENT-LEVEL AND SONG-LEVEL CLASSIFICATION.

Class	Segment-level			Song-level		
	Precision	Recall	F1	Precision	Recall	F1
blues	0.60	0.97	0.75	0.87	1.00	0.93
classical	0.85	0.99	0.92	0.95	0.95	0.95
country	0.78	0.69	0.73	0.58	0.90	0.71
disco	0.73	0.84	0.78	0.94	0.75	0.83
hiphop	0.77	0.85	0.81	0.83	0.95	0.88
jazz	0.92	0.86	0.89	0.95	0.95	0.95
metal	0.85	0.96	0.90	0.95	0.95	0.95
pop	0.87	0.56	0.68	0.94	0.85	0.89
reggae	0.94	0.49	0.64	0.93	0.70	0.80
rock	0.62	0.52	0.56	0.79	0.55	0.65
Accuracy	0.77			0.85		
Macro Avg F1	0.77			0.85		

V. CONCLUSION

This paper designs and implements an automatic music genre classification system based on Convolutional Neural Networks. Experiments show that using Log-Mel spectrograms as input, combined with a deep CNN architecture, comprehensive regularization strategies, and majority voting at the song level, can achieve high-precision music genre classification on public datasets, validating the effectiveness of deep learning for this task. The system code has been modularly implemented, providing a foundation for subsequent research and applications.

Main Contributions: A complete and reproducible engineering implementation covering the entire pipeline from feature extraction, model training to evaluation and deployment; using an effective combination of regularization strategies to significantly mitigate overfitting on limited data; employing a dual-level evaluation system, clearly distinguishing segment-level and song-level performance, and validating the role of majority voting in improving practicality; and revealing model behavior and classification bottlenecks through multi-perspective charts.

Limitations and Future Work: This study is limited by the scale and annotation noise of the GTZAN dataset. Future

work could validate model generalization on larger-scale, more finely annotated datasets. In terms of models, more advanced network architectures (such as ResNet, Transformer), multi-feature fusion (MFCC, rhythm features, etc.), and the introduction of temporal modeling modules (such as RNN/LSTM) to capture intra-song structural information could be explored, pushing models towards finer-grained and more robust music content understanding.

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